GOLDEN CAFE LTD

WHOLESALE ERP SYSTEM

Comprehensive System User Manual

A Complete Part-by-Part Guide, Auto-Calculations Logic,
and Unique Code Generation Schemas for Clients & Operations.

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Guide to the modules and calculation processes explained in this manual

Document Structure

Chapter 1: ERP Modules Overview

A high-level explanation of the system modules and architecture

System Architecture & Data Flows

The Golden Cafe Wholesale ERP is an integrated system designed to manage manufacturing, inventory, sales, procurement, payments, and human resources in a single centralized platform. Data flows seamlessly across modules to prevent discrepancies.

1. Catalog & Parts Management

Acts as the system's core registry. Contains all parts, raw materials, semi-finished goods, finished products, and packaging configurations. Every item holds cost targets, base prices, tax rules, and unit configurations.

2. Procurement & Receiving (GRN)

Manages relations with ingredient suppliers. Purchase Orders (PO) track purchase contracts, and Goods Receipt Notes (GRN) log inbound material counts, record quality rejections, and trigger dynamic production forecasts.

3. Manufacturing & MES (Manufacturing Execution System)

Executes process templates on the factory floor. Batches monitor staff counts, drying times, firewood/electricity utility loads, and powder outputs. Integrates directly with QA checkpoints to release inventory.

4. Sales, Invoicing & Logistics

Handles wholesale clients. Processes sales orders, converts them into commercial invoices, checks available stock, generates security gate passes, and logs delivery fleet trips.

Chapter 2: Step-by-Step Operations

Functional guidelines on how to use key interfaces in the ERP

1. Creating a Part / Product

- **Step A:** Navigate to "Products" and click "Add Product". The Product Code field will display "Auto-generated after category selection".
- **Step B:** Select the appropriate Category (e.g. Mechanical). The system queries the backend to generate the next unique Julian date code.
- **Step C:** Enter details including Name, Unit of Measure, base price, and applicable tax rates, then click Save.

2. Receiving Inbound Raw Materials

- **Step A:** Open a pending Purchase Order and click "Receive Goods" to launch the GRN Modal.
- **Step B:** Input the exact quantities received and any rejected volumes. The system automatically computes Accepted Qty.
- **Step C:** If a recipe conversion rule is active, a "Live Yield Forecast" card will appear under the row, displaying expected output weight.
- **Step D:** Input Supplier invoice, delivery note, vehicle parameters, and click "Confirm Receipt".

3. Logging Factory Batches

- **Step A:** Navigate to "Batches" and select "Log New Batch". Specify the raw material, supplier source, and target process template.
- **Step B:** Record drying hours, shifts, firewood consumed (Kg), and electricity consumed (kWh).
- **Step C:** Enter output weights and submit. The batch will be sent to the QC Queue to be approved for sale.

Chapter 3: Auto-Calculations Engine

The mathematical logic powering inventory valuation and projections

1. Bill of Materials (BOM) Costing

During BOM creation or updates, the system pulls standard raw material costs. It factors in wastage allowances to compute the true material cost of production.

BOM Cost Formula

```
Cost = Qty * (1 + Wastage% / 100) * UnitCost + Labor + Overhead
```

Calculates the net cost of producing an output batch, adding standard labor charges and overhead percentages.

2. Production Batch Efficiency

Calculated dynamically upon logging output weights to evaluate raw material yields and processing efficiency.

Efficiency Ratio Formula

```
Efficiency % = (Output Weight / Input Weight) * 100
```

Determines processing wastage. Higher efficiency indicates better moisture reduction or less material loss.

3. Yield & Resource Projections

When running forecasts, the backend processes past batch runs for a selected product. It maps index values (X) to efficiency percentages (Y) and computes Simple Linear Regression ($y = mx + c$) and 5-batch Moving Averages.

Linear Regression Trendline

```
Expected Output = InputWeight * (Slope * NextIndex + Intercept) / 100
```

Projects future output yields based on the mathematical slope of historical batch improvements.

Chapter 4: Unique Key Generation

The structure and logic behind the automated codes in the system

1. Parts Unique Key Format

All parts and products registered under categories use a Julian Calendar date key format. This key is generated instantly upon selecting a category and is saved as a read-only parameter.

Parts Unique Code Structure

```
P - [CategoryShortCode] - [YearShort][JulianDayOfYear] - [SequenceNo]
```

Example: P-MAC-26155-02 (Part - Mechanical - Year 2026, Day 155 - Sequence 02)

Key Parameters Explained

- **CategoryShortCode:** A dynamic 3-letter uppercase abbreviation based on category code/name (e.g. ELE = Electrical, PKG = Packaging).
- **YearShort & JulianDay:** The last two digits of the year (e.g., 2026 = 26) followed by the sequential day of the year from 001 to 365/366 (e.g., June 4th = 155). Padded to 3 digits.
- **SequenceNo:** A 2-digit counter (starting at 01) that resets daily per category, allowing multiple distinct additions per day.

2. Production Batch Code Format

Manufacturing runs use a Julian code incorporating the supplier code to trace raw materials back to their source.

Batch Unique Code Structure

```
[SupplierShortCode] - ALE [YearShort][JulianDayOfYear]
```

Example: CHAMINDA-ALE26002 (Supplier Chaminda, processed on Year 2026, Day 002)

Chapter 5: System Guidelines

Best practices for users and administrators of the system

System Security & Concurrent Locks

The ERP uses database-level atomic operations. The sequence counters in the database are incremented atomically inside a single query block. This ensures that even if 100 users create a product or receive goods at the exact same millisecond, the system guarantees 100 unique codes without overlapping.

Operating Guidelines & Data Quality

- **Keep UTC timezone stable:** The server uses UTC calculation for Julian days. Changing server timezone settings could shift the Julian Day calculation by one day.
- **Expiring Counters:** Sequence counter keys automatically expire in the database after 30 days of inactivity to keep database indices lightweight.
- **Validation Constraints:** Unique constraints are enforced at the database layer. If any network failure bypasses frontend checks, Mongoose will block duplicate codes, preventing data corruption.

Help & Support

For technical support, custom role configurations, or API integrations, contact your System Administrator or developer team. Ensure regular daily backups are enabled in Settings.